Final Benchmarks and Robustness of the Graph-Laplacian Water-Quality Model

Where the River Network Helps, and Where It Does Not

---

1 Summary

This note collects the final results of the Graph-Laplacian model since the Nov 5 meeting. The full analysis, with every table and caveat, is in `kron_vs_full_laplacian_report.pdf`.

- With fair model selection for every model, the graph model is clearly best for **specific conductance** and **pH**. It ties with the linear models for water temperature and is mid-pack for dissolved oxygen (DO) and turbidity.
- Its advantage holds when several stations are held out at once, but **not** when a whole subbasin is held out. The graph model interpolates between stations of the same subbasin.
- Most model extensions did not help. The exception is the reading at neighbouring stations at the same hour, which clearly improves pH.
- Every model fails in the Lower Colorado, whose three stations all sit below large dams.
- Training on the longer record (water years 2021–2026) does not help as it stands: the model has no year term, so water temperature comes out about 1 °C too cold.

2 The Model and the Graph

For station i and hour t the model is

$$y_{i,t} = \beta_0 + X_{i,t}\beta + S_i\gamma + g_i + \varepsilon_{i,t},$$

and the Graph-Laplacian model minimizes

$$\boxed{\sum_{i=1}^{m} \frac{1}{n_i} \sum_{t} \big(y_{i,t} - \beta_0 - X_{i,t}\beta - S_i\gamma - g_i\big)^2 \;+\; \lambda_L\, g_B^T L_{\text{eff}}\, g_B \;+\; \lambda_\beta\big(\|\beta\|_2^2 + \|\gamma\|_2^2\big)}$$

over β_0, β, γ and g_B, subject to

$$\sum_{i=1}^{m} g_i = 0.$$

Here,

- $i = 1, \ldots, m$ indexes the monitoring stations (graph nodes) in the training data, and t the hours observed at station i;

- $y_{i,t}$ is the observed water-quality value (the target) at station i and hour t;
- $X_{i,t}$ is the row of time-varying predictors: the other water-quality variables used for this target and the daily and annual Fourier terms;
- S_i is the row of static MERIT Hydro covariates of station i (elevation, log upstream area, log channel width, height above nearest drainage, local slope);
- β_0 is the intercept, and β and γ are the coefficient vectors of $X_{i,t}$ and S_i;
- g_i is the station effect of station i: the station's offset that the predictors do not explain;
- $g_B = (g_1, \ldots, g_m)^T$ is the vector of station effects at the retained monitoring nodes B;
- $\varepsilon_{i,t}$ is the residual error;
- n_i is the number of hourly rows at station i; the factor $1/n_i$ gives every station the same total influence, as in the Oct 29 notes;
- $L_{\text{eff}} = L_{BB} - L_{BI}L_{II}^{-1}L_{IB}$ is the $m \times m$ Kron-reduced Laplacian of the Nov 5 notes [1], with L the weighted Laplacian of the river network and I its interior nodes;
- $\lambda_L \geq 0$ controls how strongly the station effects are smoothed over the network; a large λ_L pulls connected stations' effects together;
- $\lambda_\beta \geq 0$ is the ridge penalty on β and γ (the intercept β_0 is not penalized), and $\|\cdot\|_2$ is the Euclidean norm;
- the constraint $\sum_i g_i = 0$ keeps the station effects from absorbing the intercept.

The Laplacian weighting, λ_L, λ_β and the feature set (which columns of $X_{i,t}$ and S_i are used) are chosen by the inner leave-one-station-out loop.

The river network is built from MERIT Hydro [2]. In the uncompressed graph every grid cell of about 90 m is a node, linked to the next cell downstream: 67,401 nodes and 67,400 edges, forming a tree (Figure 1). All but 66 of these nodes are pass-through cells with one cell upstream and one downstream; the rest are station cells, confluences, sources and the outlet.

Compression keeps only stations, confluences and river ends as nodes. Every chain of pass-through cells between two nodes becomes one edge, weighted by its length (Figure 2). The result has 67 nodes and 66 edges, of which 26 nodes hold water-quality stations.

Kron reduction is exact for this model. It matches the full 67,401-node Laplacian to rounding error, gives identical hyperparameters and R^2, and is about 150× faster.

3 Final Benchmark

All results use leave-one-station-out (LOO) prediction, as on Nov 5. Unlike the Nov 5 table, every model now chooses its features and settings in an *inner* leave-one-station-out loop on the training stations only. The held-out station is never used to choose anything.

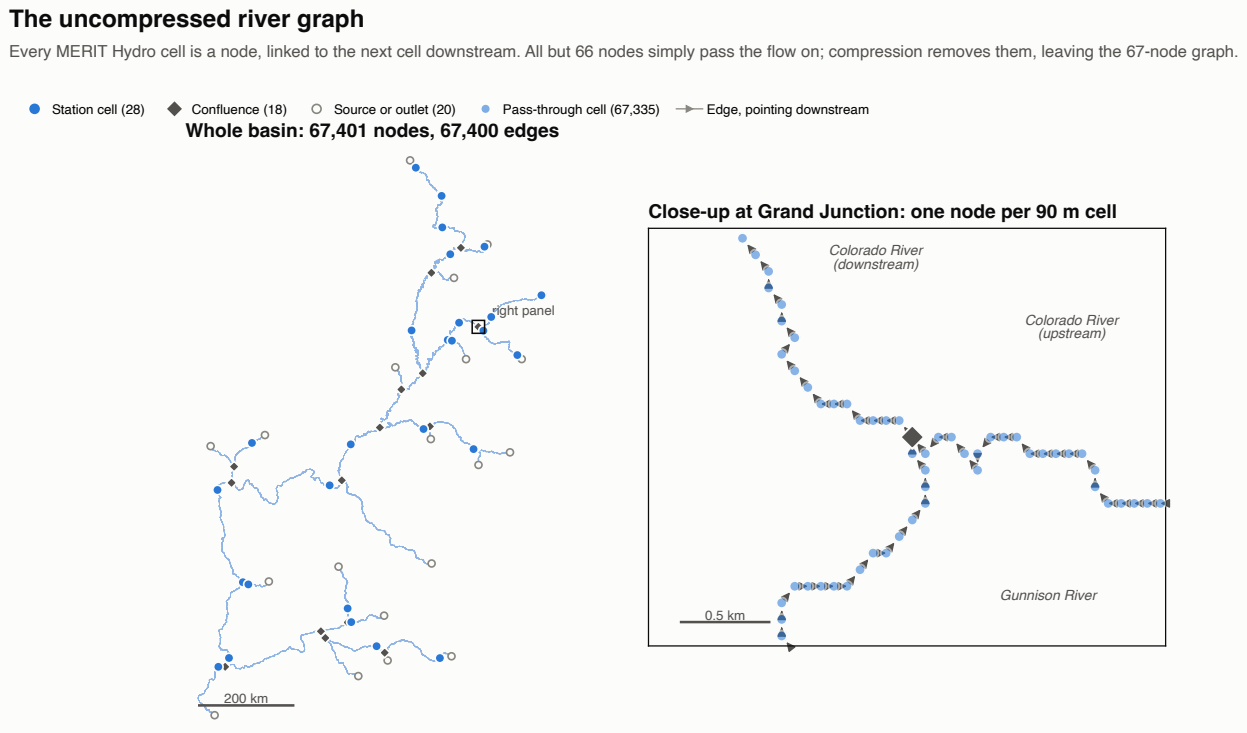


Figure 1: The uncompressed graph: the whole basin (left) and a close-up of the Gunnison confluence at Grand Junction (right), where each dot is one 90 m cell and each arrow an edge pointing downstream.

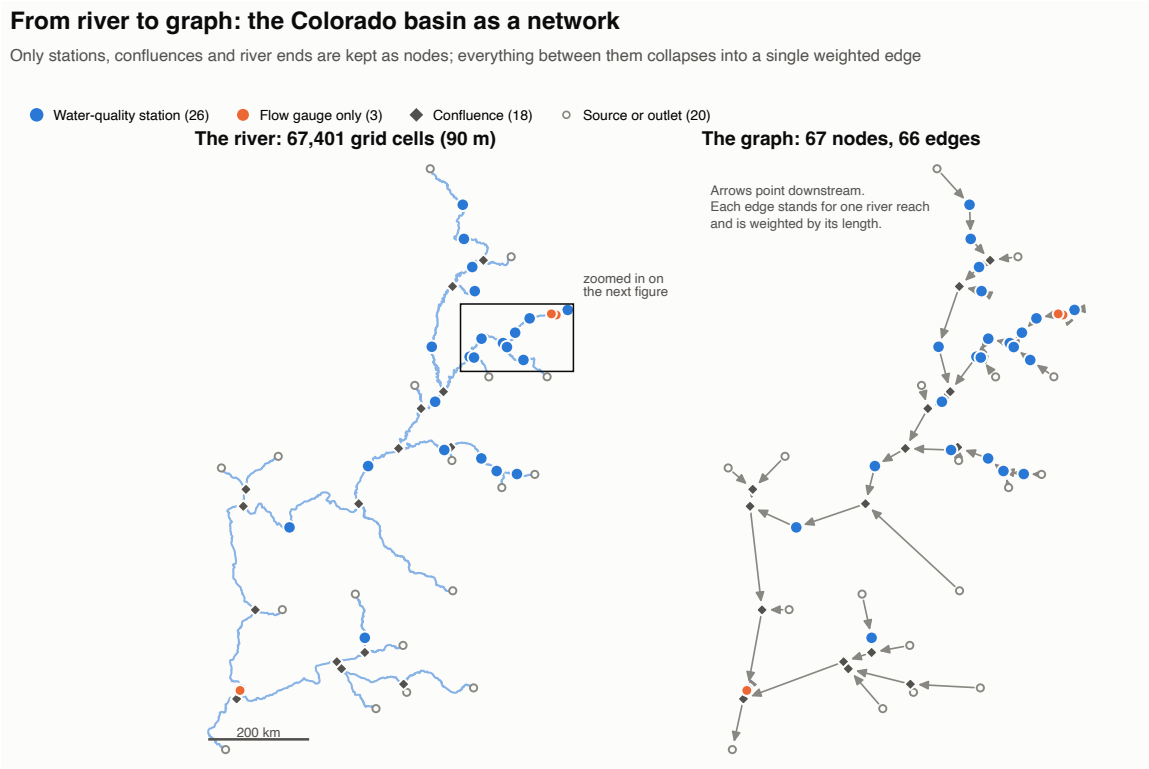


Figure 2: The river network (left) and the graph built from it (right).

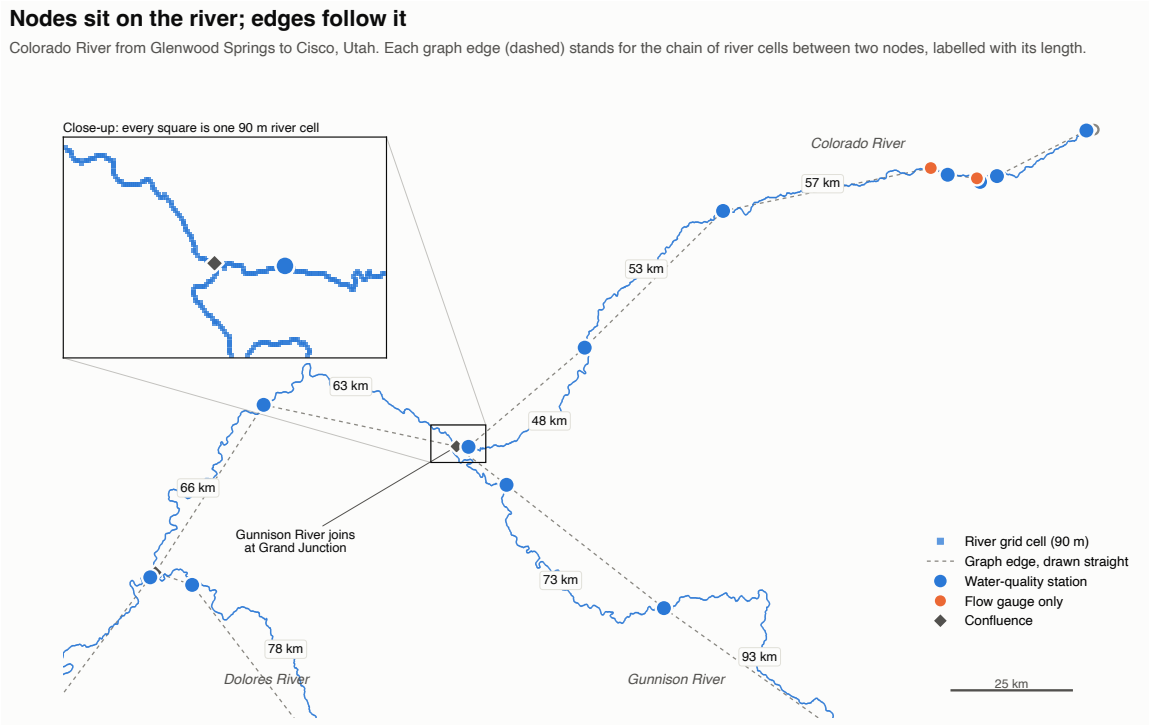


Figure 3: Close-up of the Colorado River from Glenwood Springs to Cisco, Utah. Each edge stands for the river cells between two nodes.

Table 1: Final LOO benchmark (R^2, equal station weights). Best in bold.

Model	Water temp.	DO	Spec. cond.	pH	Turbidity
Stations	24	17	24	19	17
Graph-Laplacian	0.8634	0.6996	**0.3847**	**0.1173**	0.1043
OLS	0.8634	0.6951	−0.0987	−0.0138	0.1360
Ridge	**0.8640**	0.7008	−0.1096	0.0029	0.1362
Elastic Net	0.8635	0.7048	−0.1089	0.0226	0.1362
Spline GAM	0.8382	0.7076	0.0218	0.0117	0.0910
Random Forest	0.7579	0.6845	−0.1740	−0.2685	0.0996
Gradient Boosting	0.8326	**0.7327**	−0.0939	−0.1713	**0.3911**
Dummy (mean)	−0.0327	−0.0286	−0.0623	−0.0301	−0.0271

Specific Conductance. The graph model is the only model that predicts unseen stations usefully:

$$R^2: \quad 0.022 \longrightarrow 0.385$$

from the best other model (Spline GAM) to the graph model. Fair selection first lowered the Nov 5 value (0.253) to 0.207. An unweighted Laplacian on the compressed graph (one step per reach), chosen by the inner loop, then raised it to 0.385.

pH. The graph model is the only model meaningfully above zero:

$$R^2: \quad 0.023 \longrightarrow 0.117.$$

Water Temperature and DO. Seasonality and water temperature explain most of the variation. The graph model ties with the linear models for water temperature and is within 0.033 of gradient boosting for DO.

Turbidity. Gradient boosting leads by a wide margin (0.391). No additive model, including the graph model, captures turbidity's nonlinear response.

3.1 What Changed Since Nov 5

The graph model's R^2 moved in two steps (Table 2).

Table 2: Graph-Laplacian LOO R^2 in the Nov 5 notes, with fair (inner-loop) selection of features and settings, and in the final model, which adds the unweighted compressed Laplacian as a candidate.

Target	Nov 5 notes	Fair selection	Final
Water temperature	0.8628	0.8628	0.8634
DO	0.7261	0.6998	0.6996
Specific conductance	0.2525	0.2068	**0.3847**
pH	0.1316	0.1299	0.1173
Turbidity	0.0145	0.1063	0.1043

1. **Fair selection.** On Nov 5 every model used a fixed, hand-picked feature set, partly designed with the leave-one-out results in view. Choosing the features inside the inner loop lowered the graph model's specific conductance from 0.253 to 0.207 and DO from 0.726 to 0.700. Turbidity rose (0.015 $\rightarrow$ 0.106) because the inner loop could choose richer seasonal terms, which helped almost every model.

2. **The unweighted network.** The final model adds one more candidate weighting: every reach of the compressed graph counts as one step, whatever its length. The inner loop chose it in all 24 folds for specific conductance, raising R^2 from 0.207 to 0.385. pH kept mostly the distance weighting, and the other targets changed by less than 0.002.

On the compressed graph every edge is a whole reach between stations, confluences and river ends. Counting reaches rather than kilometres says that two stations are similar when few confluences separate them. A plausible reason this suits salinity, not tested here, is that it changes mostly where tributaries of different chemistry join.

3.2 What the Graph Term Contributes

Removing one variable at a time from the model shows what each target depends on (Table 3).

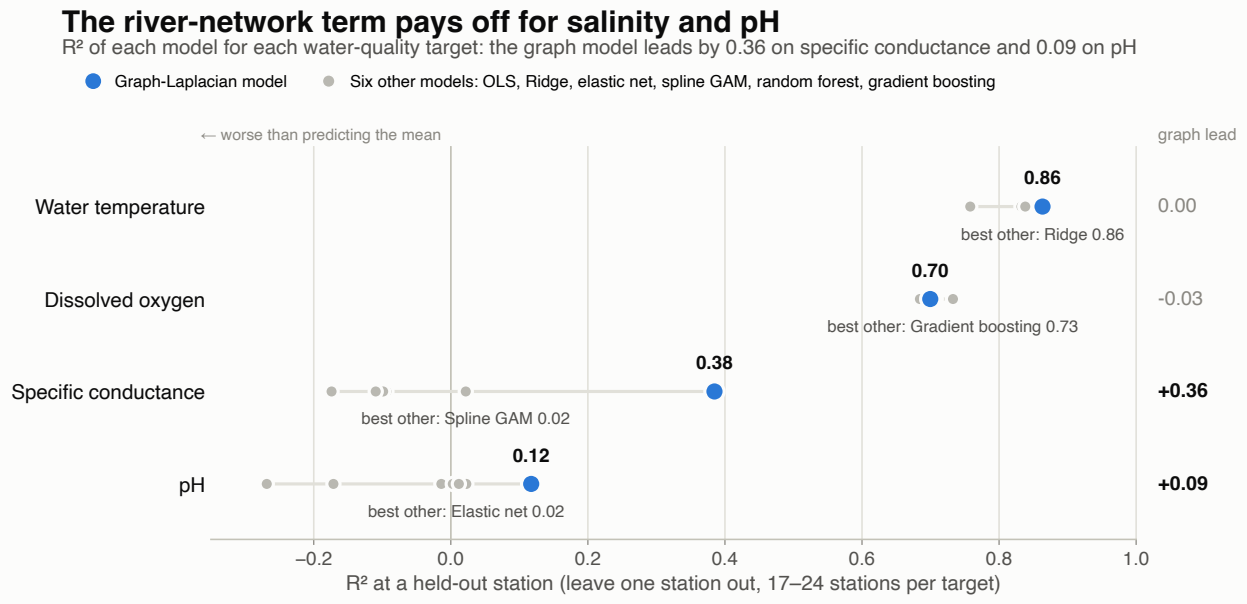


Figure 4: LOO R^2 of the graph model (blue) and the six other models (grey).

Table 3: Ablation: change in LOO R^2 when one variable is removed from the graph model (a separate run with each target's fixed chemistry predictors; the full-model R^2 is in the first row). Large negative values are variables nothing else can replace.

Variable removed	Water temp.	DO	Spec. cond.	pH
Full model	*0.860*	*0.722*	*0.315*	*0.131*
Graph term	0.000	−0.008	**−0.449**	**−0.109**
Annual cycle	**−0.721**	−0.010	−0.107	−0.035
Daily cycle	−0.009	−0.116	+0.003	−0.043
Water temperature	–	**−0.211**	+0.025	–
Elevation	−0.030	−0.032	−0.042	0.000
Local slope	−0.039	−0.006	−0.150	−0.026

- The **graph term** is the most valuable input for specific conductance and pH, and adds nothing for water temperature and DO.
- **Water temperature** is driven by the annual cycle, and **DO** by water temperature and the daily cycle (photosynthesis).
- Among the MERIT Hydro covariates only local slope consistently helps.

4 Extensions That Did Not Help

Each extension was offered to the inner loop as an extra option and compared with the final model on identical rows (Table 4).

Table 4: Change in the graph model's LOO R^2 from each extension. With 17–24 stations, changes below about 0.03 are noise. The station-bootstrap 95% interval excludes zero only where marked *.

Extension	Water temp.	DO	Spec. cond.	pH
Other weights and settings	+0.000	+0.003	−0.034	+0.008
Dams as resistances	−0.001	+0.014*	0.000	+0.006
Tail-up covariance	0.000	−0.007	−0.227	+0.004
REML settings	0.000	+0.015	−0.153*	−0.115*
Network-varying annual cycle	−0.022	−0.015	−0.005	−0.002
Upstream geology, crop land	0.000	−0.002	−0.019	0.000
Graph + gradient boosting	−0.018	−0.009	−0.015	+0.051

- **Other edge weights and settings** (confluence counts, length powers, log target, wider λ grid): no gain; specific conductance −0.034.
- **Dams as high-resistance edges**: no gain for water temperature or turbidity.
- **Flow-directed (tail-up) covariance** [3]: specific conductance −0.23.
- **REML** choice of settings: worse for specific conductance and pH.
- **Discharge** as a predictor: no reliable change for the graph model.
- **Upstream-watershed geology and crop land** (EPA StreamCat [4]): specific conductance −0.019. The network already carries this information.
- **Graph + gradient boosting hybrid**: raised turbidity to 0.349 under LOO, but on a later time period it was never better than the graph model alone and clearly worse for pH, so it is not adopted.

5 Robustness to the Evaluation Split

LOO holds out one station at a time, so a test station's neighbours are always in training. Three further splits test how much the results depend on that.

- **3 stations out**: 20 random draws of 20% of the stations, tested together.
- **Subbasin out**: each of five subbasins (Colorado headwaters, Gunnison, Green, San Juan, Lower Colorado) held out whole.
- **Later period**: trained before 2026-01-01, tested on the other stations after it.

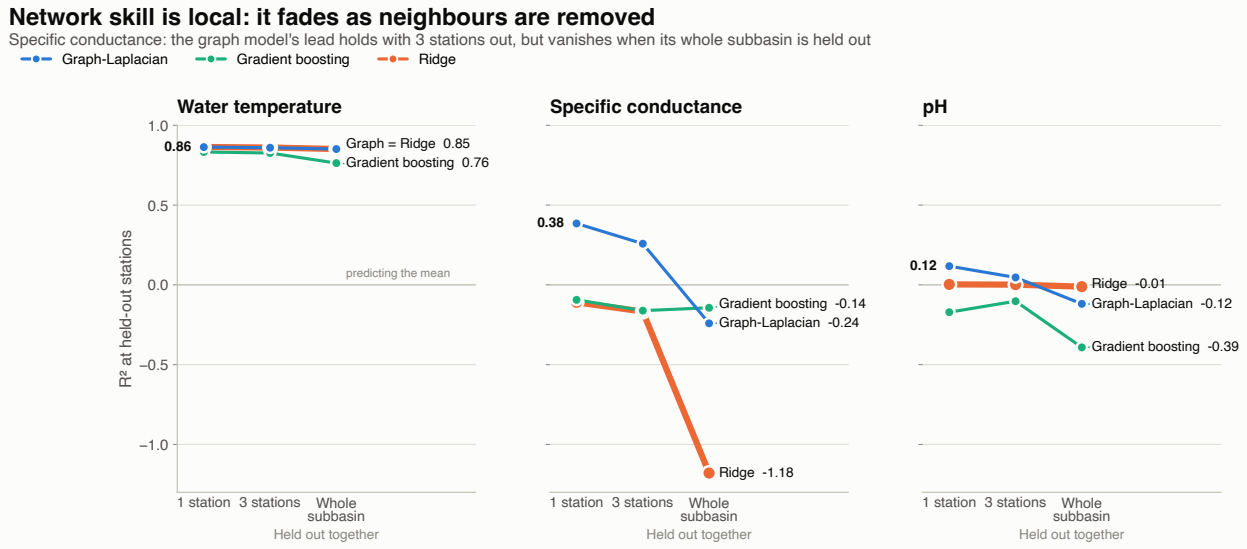


Figure 5: R^2 as more of the network is held out.

Table 5: R^2 of the graph model, with the best other model in parentheses. The later-period run compared only Ridge and gradient boosting with the graph model.

Split	Water temp.	DO	Spec. cond.	pH
LOO	0.863 (0.864)	0.700 (0.733)	**0.385** (0.022)	**0.117** (0.023)
3 stations out	0.859 (0.861)	0.702 (0.746)	**0.258** (−0.060)	**0.047** (0.010)
Subbasin out	0.851 (0.852)	0.623 (0.709)	−0.241 (−0.144)	−0.119 (−0.011)
Later period	0.774 (0.760)	0.543 (0.683)	**0.089** (−0.116)	−0.136 (−0.022)

Water temperature transfers well under every split: it follows climate and season, which all subbasins share.

Specific conductance keeps its lead with three stations out and on a later period, but

$$R^2: \quad 0.385 \longrightarrow -0.241$$

when the whole subbasin is held out. The graph model then does worse than predicting the training mean. Its skill is therefore interpolation from other stations of the same subbasin. It does not carry a salinity level into a subbasin with no monitored station.

pH follows the same pattern on a smaller scale. On a later period every model is below zero.

6 Where in the Basin the Model Works

Grouping the LOO predictions by subbasin shows where each model works (Tables 6–9 and Figure 6). Within a subbasin, R^2 is measured against that subbasin's own mean, so it is

lower than the pooled values. The best of the three models is in bold; ties at two decimals are broken by the unrounded value.

Table 6: LOO R^2 for water temperature within each subbasin.

Subbasin	Stations	Graph	Ridge	Grad. boosting
Colorado headwaters	7	0.94	0.94	**0.95**
Green	6	0.86	0.86	**0.86**
Gunnison + Colorado–Dolores	5	0.88	0.88	**0.95**
San Juan	3	0.87	0.87	**0.87**
Lower Colorado + Verde	3	0.20	**0.20**	−0.65

Table 7: LOO R^2 for DO within each subbasin.

Subbasin	Stations	Graph	Ridge	Grad. boosting
Colorado headwaters	7	0.75	0.76	**0.77**
Green	2	0.86	0.81	**0.95**
Gunnison + Colorado–Dolores	4	0.83	0.82	**0.86**
San Juan	1	0.76	0.89	**0.94**
Lower Colorado + Verde	3	**−0.37**	−0.41	−0.45

Table 8: LOO R^2 for specific conductance within each subbasin.

Subbasin	Stations	Graph	Ridge	Grad. boosting
Colorado headwaters	7	**0.52**	−0.17	−0.19
Green	6	**0.37**	−0.88	−1.98
Gunnison + Colorado–Dolores	5	**0.11**	−0.30	−0.23
San Juan	3	**−0.79**	−6.99	−5.14
Lower Colorado + Verde	3	**−0.52**	−1.03	−0.72

Table 9: LOO R^2 for pH within each subbasin.

Subbasin	Stations	Graph	Ridge	Grad. boosting
Colorado headwaters	6	**0.16**	−0.00	−0.07
Green	2	**−0.06**	−0.57	−1.18
Gunnison + Colorado–Dolores	4	**0.15**	−0.08	−0.08
San Juan	4	−0.08	−0.05	**0.30**
Lower Colorado + Verde	3	**−0.27**	−0.40	−2.70

- **Water temperature** is predicted well (0.86–0.95) in every subbasin except the Lower Colorado. The three models are within 0.07 of each other.

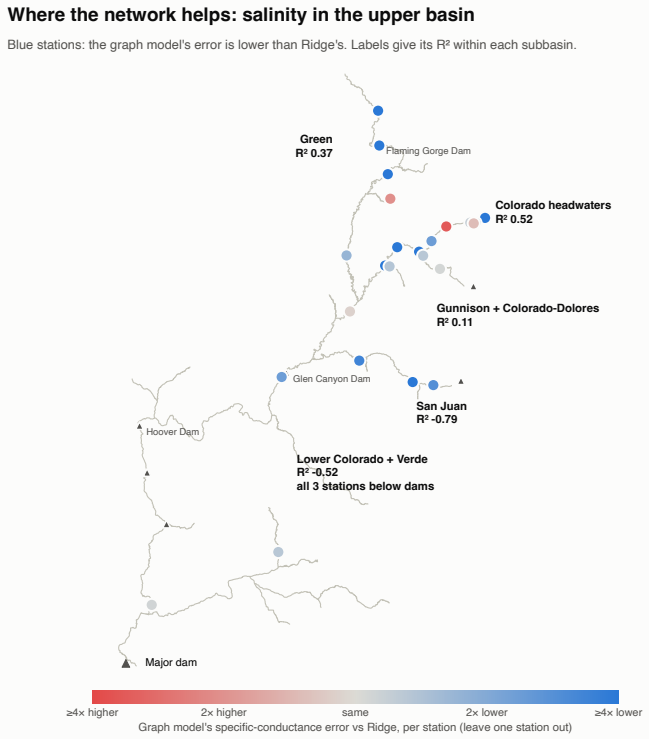


Figure 6: Graph-model specific conductance error relative to Ridge at each station (blue: lower).

- **DO**: gradient boosting is best in four of five subbasins, by 0.01 to 0.18 over the graph model (the largest gap at the single San Juan station).
- **Specific conductance**: only the graph model is positive in any of these subbasins, all three in the upper basin, where Ridge and gradient boosting are negative. Holding a subbasin out removes exactly these neighbours (Section 5).
- **pH**: the graph model is best in four of five subbasins. Gradient boosting is best in the San Juan (0.30).

The **Lower Colorado** fails for every target and every model, even water temperature (best 0.20). Its three stations are all tailwaters of large dams (Lees Ferry below Glen Canyon Dam, the Verde below Bartlett Dam, and above Imperial Dam). Releases of deep reservoir water mute the annual cycle that the seasonal terms expect: the median within-record standard deviation of water temperature is 2.5 °C there, against 5.7–6.5 °C elsewhere.

6.1 Upper-Basin Stations Only

Table 10 scores the same LOO predictions on subsets of the stations. The models are not refitted. Each R^2 is pooled over the stations of the subset, with equal station weights.

Table 10: LOO R^2 of the graph model on station subsets, with the best other model in parentheses. Station counts in brackets. The Upper Colorado Basin is every subbasin above Lees Ferry, i.e. all but the Lower Colorado.

Stations included	Water temp.	DO	Spec. cond.	pH
All stations	0.863 (0.864) [24]	0.700 (0.733) [17]	**0.385** (0.022) [24]	**0.117** (0.023) [19]
Excl. San Juan and Lower Colorado	0.900 (0.923) [18]	0.796 (0.831) [13]	**0.381** (−0.003) [18]	**0.216** (0.082) [12]
Upper Colorado Basin	0.899 (0.918) [21]	0.795 (0.840) [14]	**0.425** (0.027) [21]	**0.140** (0.076) [16]
Colorado headwaters only	0.937 (0.952) [7]	0.753 (0.775) [7]	**0.521** (0.269) [7]	**0.161** (0.056) [6]
Lower Colorado only	0.202 (0.203) [3]	−0.368 (−0.407) [3]	−0.520 (−0.107) [3]	−0.273 (−0.397) [3]

- **Specific conductance and pH**: the graph model stays clearly first in every subset except the Lower Colorado, reaching 0.42–0.52 and 0.14–0.22 in the upper basin.
- **Water temperature and DO** rise for every model once the Lower Colorado tailwaters are left out (0.86 → 0.90 and 0.70 → 0.80). Gradient boosting then leads by 0.02–0.05.
- Leaving out the San Juan barely changes specific conductance (0.385 → 0.381), although the graph model is negative within it. Its stations widen the spread between stations, which the graph model largely captures.
- **Lower Colorado only**: every model is below zero except for water temperature (about 0.20), and none beats the training mean for specific conductance (−0.088). The graph model is the least poor for DO and pH.
- These subsets were chosen after seeing the results, so they describe where the model works; they are not an independent test. A fair upper-basin-only result would need the models refitted and validated on those stations alone.

7 Neighbouring Stations Help pH

One extension did help: the anomaly of the three nearest training stations at the same hour (Table 11).

Table 11: Change in LOO R^2 when the neighbour reading is offered to the inner loop (paired, same rows). *: the station-bootstrap 95% interval excludes zero. Last column: pH R^2 with it.

Model	Water temp.	DO	Spec. cond.	pH	pH R^2
Graph-Laplacian	+0.010	+0.012	+0.007	+0.061	**0.178**
OLS	+0.011	+0.006	+0.059*	+0.143*	0.130
Ridge	+0.011	+0.004	+0.049*	+0.118*	0.121
Elastic Net	+0.011	+0.016	+0.049*	+0.121*	0.143
Spline GAM	+0.004	−0.009	−0.002	+0.141*	0.153

- **pH**: every conventional model gains 0.12–0.14 (intervals above zero), and the graph model rises from 0.117 to 0.178, the best LOO pH so far.

- **Specific conductance**: the linear models gain about 0.05; the graph model barely changes.
- Water temperature and DO gain about 0.01; turbidity falls slightly.

This feature needs live readings from neighbouring stations, so it answers a different question: filling in an unmonitored site while its neighbours report.

8 Longer Training History

The benchmark uses only part of each station's record: 85% of the water-temperature test rows are from 2025–26. We downloaded every station back to October 2020 (water year 2021), which adds about 5× the training rows. The held-out station is still tested on exactly the same rows, and its own added rows are never used, so the comparison is paired (Table 12).

Table 12: LOO R^2 trained on the current rows → with the 2021–2026 history (same test rows). *: the paired station-bootstrap 95% interval of the change excludes zero.

Model	Water temp.	DO	Spec. cond.	pH	Turbidity
Graph-Laplacian	.863 → .816*	.700 → .699	**.385 → .334**	.117 → −.037*	.104 → .043
OLS	.863 → .840*	.695 → .702	−.099 → −.022*	−.014 → −.011	.136 → .032
Ridge	.864 → .840*	.701 → .708	−.110 → −.032*	.003 → .013	.136 → .031
Elastic Net	.863 → .841*	.705 → .712	−.109 → −.039*	.023 → .029	.136 → .033
Spline GAM	.838 → .745	.708 → .719	.022 → .055	.012 → .009	.091 → .085

- **Water temperature gets worse for every model.** Almost all of the loss is a shift in level: the models now predict the held-out stations about 1 °C too cold, while the error within each station is unchanged. The model has no year term, so it learns the 2021–26 average, but the test rows are mostly from 2025–26, and water year 2026 was 1.2 °C warmer than usual (2023 was 1.1 °C colder). Removing one common offset brings Ridge back to 0.860.
- **pH: the graph model loses its lead** (0.117 → −0.037), because the inner loop now chooses much heavier shrinkage. The other models do not change.
- **Specific conductance:** the linear models gain about 0.08 but stay below zero; the graph model falls slightly and keeps a large lead (0.334 against 0.055).
- DO gains less than 0.01; turbidity falls for the linear models.
- **Upper-basin stations only** (the same predictions, scored without the Lower Colorado): the same pattern. Water temperature falls by about 0.025 for every model (0.899 → 0.876 for the graph model), specific conductance 0.425 → 0.391 (Ridge −0.07 → 0.02), pH 0.140 → 0.028, now behind the Spline GAM (0.087). In the headwaters alone the offset is small and water temperature holds at 0.935.

The longer record should be retried with a water-year effect estimated from the training stations, which are observed at the same times as the held-out station.

9 Conclusions and Next Steps

- The Graph-Laplacian model is the model to use for **specific conductance** and **pH** at unmonitored sites, provided the subbasin has monitored stations.
- For water temperature and DO, simple linear models are just as good. For turbidity, gradient boosting is best.
- Claims should be limited to interpolation within monitored subbasins. Predicting an unmonitored subbasin is not supported.
- The longer record (back to October 2020) does not help without a year term.
- **Running:** gradient boosting with the neighbour reading.
- **Proposed:** the longer record with a water-year effect, with turbidity clipped at the sensor maximum, also retrained on the upper basin only.
- **Proposed:** a dam-tailwater indicator, or a weaker seasonal cycle below major dams, for the Lower Colorado stations.

References

[1] F. Dörfler and F. Bullo, "Kron Reduction of Graphs With Applications to Electrical Networks," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 60, no. 1, pp. 150–163, 2013, doi: 10.1109/TCSI.2012.2215780.

[2] D. Yamazaki, D. Ikeshima, J. Sosa, P. D. Bates, G. H. Allen, and T. M. Pavelsky, "MERIT Hydro: A High-Resolution Global Hydrography Map Based on Latest Topography Dataset," *Water Resources Research*, vol. 55, pp. 5053–5073, 2019.

[3] J. M. Ver Hoef, E. Peterson, and D. Theobald, "Spatial Statistical Models That Use Flow and Stream Distance," *Environmental and Ecological Statistics*, vol. 13, no. 4, pp. 449–464, 2006.

[4] R. A. Hill, M. H. Weber, S. G. Leibowitz, A. R. Olsen, and D. J. Thornbrugh, "The Stream-Catchment (StreamCat) Dataset: A Database of Watershed Metrics for the Conterminous United States," *Journal of the American Water Resources Association*, vol. 52, no. 1, pp. 120–128, 2016.